Quiz 3 Formula Sheet

1 Expectation and Variance

1.1 Expectation of a Function

For a discrete random variable X with PMF $p_X(x) = P(X = x)$,

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x).$$

In particular,

$$\mathbb{E}[X] = \sum_x x\, p_X(x).$$

1.2 Linearity of Expectation and Independent Products

For any random variables X, Y and constants a, b, c,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y],$$

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

If X and Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

1.3 Variance and Standard Deviation

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}, \qquad \mathbb{E}[X^2] = \mathrm{Var}(X) + \big(\mathbb{E}[X]\big)^2.$$

For constants a, b,

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X),$$

$$\mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X).$$

If X and Y are independent,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

1.4 Sums of Identically Distributed Indicators

Let

$$X = I_1 + \cdots + I_n, \qquad p_1 = \mathbb{E}[I_1], \qquad p_{12} = \mathbb{E}[I_1 I_2].$$

If the indicators are identically distributed ($p_1 = p_2 = ... = p_n$), then

$$\mathbb{E}[X] = np_1,$$

$$\mathrm{Var}(X) = np_1 + n(n-1)p_{12} - (np_1)^2.$$

If the indicators are also independent, then $p_{12} = p_1^2$ and

$$\mathrm{Var}(X) = np_1(1 - p_1).$$

2 Probability Bounds

2.1 Markov's Inequality

If X is a non-negative random variable, $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

2.2 Chebyshev's Inequality

For any random variable X and $k > 0$,

$$P(|X - \mathbb{E}[X]| \geq k \,\mathrm{SD}(X)) \leq \frac{1}{k^2}$$

$$P(|X - \mathbb{E}[X]| \geq k) \leq \frac{(\mathrm{SD}(X))^2}{k^2}$$

3 Discrete Distributions

3.1 Binomial Distribution

If X counts successes in n independent trials with success probability p, then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

3.2 Geometric Distribution

If X is the number of independent trials up to and including the first success, then

$$P(X = k) = (1-p)^{k-1} p, \qquad k = 1, 2, \ldots.$$

The tail probability is

$$P(X > k) = (1-p)^k.$$

Also,

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

3.3 Memorylessness of the Geometric Distribution

The geometric distribution is memoryless:

$$P(X = j + k \mid X > j) = P(X = k), \qquad j, k = 1, 2, \ldots.$$

Conversely, any memoryless random variable supported on $\{1, 2, 3, \ldots\}$ has a geometric distribution.

4 More Discrete Distributions

4.1 Hypergeometric Distribution

Suppose n objects are sampled uniformly without replacement from N objects, G of which are good. If X is the number of good objects selected, then

$$P(X = k) = \frac{\binom{G}{k}\binom{N-G}{n-k}}{\binom{N}{n}}.$$

The expectation is

$$\mathbb{E}[X] = n\frac{G}{N}.$$

With $p = G/N$, the variance is

$$\mathrm{Var}(X) = np(1-p)\frac{N-n}{N-1}.$$

4.2 Poisson Distribution

If $X \sim \mathrm{Poisson}(\mu)$, then

$$P(X = k) = e^{-\mu}\frac{\mu^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \mu.$$

For a nonnegative integer m,

$$P(X > m) = 1 - \sum_{k=0}^{m} e^{-\mu}\frac{\mu^k}{k!}.$$

4.3 Poisson Random Scatter

For a Poisson random scatter with constant intensity $\lambda > 0$, the number of points in an interval of length t satisfies

$$N(t) \sim \mathrm{Poisson}(\lambda t).$$

4.4 Negative Binomial Distribution

Let Y be the number of failures before the rth success in independent Bernoulli(p) trials. Then $Y \sim \mathrm{NegBin}(r, p)$ and

$$P(Y = n) = \binom{n+r-1}{r-1}p^r(1-p)^n, \qquad n = 0, 1, 2, \ldots.$$

$$\mathbb{E}[Y] = \frac{r(1-p)}{p}, \qquad \mathrm{Var}(Y) = \frac{r(1-p)}{p^2}.$$

Notes

1. Roll a die until there is a duplicate. Let $X = \#$ of trials *before* the duplicate. Find:
   Example: 3 1 4 **1** $(k = 6, x = 3)$

a) $P(X = x)$
b) $P(X \geq x)$
c) $E(X)$ as a summation

a) First x rolls must be distinct, roll $x+1$ must match one of the x values

$$P(X=x) = \frac{k(k-1)\cdots(k-x+1)}{k^x} \times \frac{x}{k} = \frac{k!}{(k-x)!} \cdot \frac{x}{k^{x+1}}$$

b) No duplicate in the first x rolls

$$P(X \geq x) = \frac{k(k-1)\cdots(k-x+1)}{k^x} = \frac{k!}{(k-x)!\,k^x}$$

c) $$E(X) = \sum_{x=1}^{k} x\,P(X=x) = \sum_{x=1}^{k} \frac{x^2\,k!}{(k-x)!\,k^{x+1}}$$

2. Suppose $X_i \sim iid\ Unif(1, \dots, 11)$.

a) Find $E(4X_1)$ and $Var(4X_1)$.

b) Find $E(X_1 + X_2 + X_3 + X_4)$ and $Var(X_1 + X_2 + X_3 + X_4)$.

$n = 11$

$E(X) = \frac{n+1}{2} = 6$

$Var(X) = \frac{n^2-1}{12} = 10$

a) $E(4X_1) = 4 \times 6 = 24$

$Var(4X_1) = 4^2 \times 10 = 160$

b) $E(X_1 + X_2 + X_3 + X_4) = 4 \times 6 = 24$

$Var(X_1 + X_2 + X_3 + X_4) = 4 \times 10 = 40$

since X_i are independent

3. Suppose $N \sim Pois(\lambda)$ and $X|N = n \sim Bin(n, p)$.

$$P(X=k) = \sum_{n=k}^{\infty} \underbrace{\frac{e^{-\lambda}\lambda^n}{n!}}_{P(N=n)} \underbrace{\binom{n}{x} p^k (1-p)^{n-k}}_{P(X=k \mid N=n)}$$

$$= \sum_{n=k}^{\infty} \frac{e^{-\lambda}(\lambda^k)(\lambda^{n-k})}{n!} \cdot \frac{n!}{k!(n-k)!} \cdot p^k (1-p)^{n-k}$$

$$= \frac{e^{-\lambda}(\lambda p)^k}{k!} \sum_{n=k}^{\infty} \frac{(\lambda(1-p))^{n-k}}{(n-k)!}$$

$$= \frac{e^{-\lambda}(\lambda p)^k}{k!} e^{\lambda(1-p)}$$

$$= \frac{e^{-\lambda p}(\lambda p)^k}{k!}$$

$X \sim Poisson(\lambda p)$

$E(X) = E[E(X|N)] = E(Np) = \lambda p$ matches Poisson

$Var(X) = E[Np(1-p)] + Var(Np) = \lambda p(1-p) + p^2\lambda = \lambda p$

Formulas:

$E(X) = E[E(X|N)]$

$= E(Np)$

$= p \cdot E(N)$

$Var(X) = E[Var(X|N)] + Var(E(X|N))$

$= p(1-p) \cdot E(N) + p^2 \cdot Var(N)$

4. Suppose $N \sim Geom(p_1)$ on $\{0, 1, \ldots\}$ and $X|N = n \sim Bin(n, p_2)$.

$$P(X=k) = \sum_{n=k}^{\infty} p_1 (1-p_1)^n \binom{n}{k} p_2^k (1-p_2)^{n-k}$$

$$= p_1 ((1-p_1)p_2)^k \sum_{n=k}^{\infty} \binom{n}{k} ((1-p_1)(1-p_2))^{n-k}$$

$$= \frac{p_1}{1-(1-p_1)(1-p_2)} \left(\frac{(1-p_1)p_2}{1-(1-p_1)(1-p_2)}\right)^k$$

$$X \sim \text{Geom}\underbrace{\left(\frac{p_1}{1-(1-p_1)(1-p_2)}\right)}_{r} \text{ on } \{0, 1, \ldots\}$$

$$E[X] = \frac{1-r}{r} = \frac{(1-p_1)p_2}{p_1}$$

$$\text{Var}(X) = \frac{1-r}{r^2} = \frac{(1-p_1)(p_2)(p_1+p_2-p_1p_2)}{p_1^2}$$

5. There are $n = 10$ parking spaces in a line assuming $n \geq 4$. Park $k = 4$ cars and let $X = \#$ of adjacent car pairs. Find:

a) $P(X = 3), P(X = 0), P(X = 1), P(X = 2)$

Examples

	Position									
x	1	2	3	4	5	6	7	8	9	10
$x = 0$	✓			✓		✓				✓
$x = 1$			✓	✓		✓				✓
$x = 2$			✓	✓					✓	✓
$x = 2$			✓	✓	✓					✓
$x = 3$			✓	✓	✓	✓				

b) $E(X)$ using indicators

c) $Var(X)$ using indicators

a) $\binom{10}{4} = 210$ equally likely configuration

x	$b = 4 - x$ blocks of cars	ordered block sizes	arrangement	Probability
0	4	$(1,1,1,1)$ → 1 way	$\binom{7}{4} = 35$	$\frac{35}{210} = \frac{1}{6}$
1	3	$(2,1,1)$ in any order → 3 ways	$3\binom{7}{3} = 105$	$\frac{105}{210} = \frac{1}{2}$
2	2	$(1,3), (2,2), (3,1)$ → 3 ways	$3\binom{7}{1} = 63$	$\frac{63}{210} = \frac{3}{10}$
3	1	(4) → 1 way	$\binom{7}{7} = 7$	$\frac{7}{210} = \frac{1}{30}$

b) Let $I_i = 1$ if spots i and $i+1$ are occupied

$$E[I_i] = P(I_i = 1) = \frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15}$$

$$E[X] = 9 \cdot E[I_i] = \frac{6}{5}$$

c) Same indicator $j = i$ (9 pairs): $I_i^2 = I_i$, so $E[I_i I_j] = \frac{2}{15}$

Neighbors, $|i-j| = 1$ (16 ordered pairs): $P = \frac{4}{10} \cdot \frac{3}{9} \cdot \frac{2}{8} = \frac{1}{30}$

Far apart, $|i-j| \geq 2$ (56 ordered pairs): $P = \frac{1}{\binom{10}{4}} = \frac{1}{210}$

$$E[X^2] = \sum_i \sum_j E[I_i I_j] = 9 \cdot \frac{2}{15} + 16 \cdot \frac{1}{30} + 56 \cdot \frac{1}{210} = 2$$

$$Var(X) = E[X^2] - (E[X])^2 = 2 - \left(\frac{6}{5}\right)^2 = \frac{14}{25}$$

Stat 134 Quiz 3 Study Guide

Retake Edition

1. Routine

1. **Name the random variable:** "Let X = the number of _____." If you can't finish the sentence, you can't pick a model.
2. **Ask about one trial.** What is a success (the thing you count or wait for, even if it is bad news)? Is the number of trials fixed (count successes: Binomial) or do you stop at the first success (wait: Geometric)? Or is there just a rate over time (Poisson)?
3. **Rewrite the English event as an event about** X using the dictionary below. **For integer variables, do this before touching any formula.**
4. **Match to the sheet,** then compute.
5. **Before submitting:** reread the full answer choice (signs, limits, e^{+15} vs. e^{-15}).

What the story gives you	Model	Sheet
fixed n, independent, count successes	Binomial(n, p)	§3.1
stop at first success, X = trial number	Geometric(p)	§3.2–3.3
steady rate per unit time or space	Poisson(λt)	§4.2–4.3
only mean and SD, asked for a "bound"	Markov / Chebyshev	§2
combine or transform variables	expectation and variance rules	§1

2. Dictionary (integer-valued X)

The words	The math
"at least k"	$X \geq k$
"more than k" / "exceeds"	$X > k$, which is $X \geq k+1$
"at most k" / "no more than"	$X \leq k$
"fewer than k"	$X \leq k-1$
"at least one"	$1 - \mathbb{P}(X = 0)$
"takes more than k trials"	first k trials all fail: $\mathbb{P}(X > k) = (1-p)^k$
"first success on trial k"	$\mathbb{P}(X = k) = (1-p)^{k-1}p$
"does not advance in the first 2 turns"	both turns fail: $(1-p)^2$
"given it already took j trials without success"	memoryless: restart the clock
"in the next 20 minutes" (rate per hour)	$t = \frac{1}{3}$, $\mu = \lambda t$
"within k of the mean"	use the complement of Chebyshev
"enter -1 if not enough information"	only if a needed fact is truly missing

3. Moves

1. **Complement.** "At least one" and "more than" are usually easier as 1 − (the opposite).
2. **Success = whatever you're tracking.** If the story waits for a miss, $p = \mathbb{P}(\text{miss})$.
3. **Reuse a computed probability as p.** If part (a) finds $p = \mathbb{P}(\text{success on one turn})$, then "repeat n times, count successes" is $\text{Bin}(n, p)$ and "how long until the first" is $\text{Geom}(p)$.
4. **Second moments.** $\mathbb{E}[W^2] = \text{Var}(W) + (\mathbb{E}[W])^2$.
5. **Functions of independent variables are independent.** If X, Y are independent, so are X^2, Y^2, hence $\mathbb{E}[X^2Y^2] = \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$.
6. **Linear change.** $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$, $\text{Var}(aX + b) = a^2\,\text{Var}(X)$. Subtraction still adds variances (if independent). $2X$ is the same X twice, unlike $X + X'$.
7. **Chebyshev needs a deviation.** Convert the event to $|X - \mathbb{E}[X]| \geq k$, find k, then use $\text{Var}(X)/k^2$. It is a bound, never an exact probability.

Key identity (Q1). For independent X, Y:

$$\text{Var}(XY) = \mathbb{E}[X^2]\,\mathbb{E}[Y^2] - \big(\mathbb{E}[X]\,\mathbb{E}[Y]\big)^2, \qquad \mathbb{E}[W^2] = \text{Var}(W) + (\mathbb{E}[W])^2.$$

4. Problems

A. Build it from the sheet (this is where the real Q1 was)

A1. Variance of a product [catch]

X has mean 2 and variance 3. Y has mean 1 and variance 5. X, Y independent. Find $\text{Var}(XY)$.

Translate: $\text{Var}(XY) = \mathbb{E}[X^2Y^2] - (\mathbb{E}[XY])^2$. Independence gives $\mathbb{E}[X^2Y^2] = \mathbb{E}[X^2]\mathbb{E}[Y^2]$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. $\mathbb{E}[X^2] = 3 + 4 = 7$, $\mathbb{E}[Y^2] = 5 + 1 = 6$. So $\text{Var}(XY) = 7 \cdot 6 - (2 \cdot 1)^2 = 42 - 4 = \mathbf{38}$.

Caveat: Independence is *given*, so there is enough information. "Not enough information" is wrong here.

A2. Mixed models

$X \sim \text{Bin}(10, 0.3)$ and $Y \sim \text{Poisson}(2)$, independent. Find $\mathbb{E}[XY]$ and $\text{Var}(XY)$.

Translate: $\mathbb{E}[X] = 3$, $\text{Var}(X) = 2.1$, so $\mathbb{E}[X^2] = 11.1$. $\mathbb{E}[Y] = 2$, $\text{Var}(Y) = 2$, so $\mathbb{E}[Y^2] = 6$. $\mathbb{E}[XY] = 6$. $\text{Var}(XY) = 11.1 \cdot 6 - 6^2 = 66.6 - 36 = \mathbf{30.6}$.

Caveat: Get each mean and variance from its own model first, then plug into the same recipe.

A3. Linear combination

X, Y independent, $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$. Find $\text{Var}(3X - 2Y + 5)$.

Translate: $9 \cdot 4 + 4 \cdot 9 = \mathbf{72}$.

Caveat: The constant drops and the minus becomes a plus.

A4. Not enough information [catch]

$\mathbb{E}[X] = 3$, $\mathbb{E}[Y] = 4$, and the problem says nothing about independence. Find $\mathbb{E}[X+Y]$ and $\mathbb{E}[XY]$.

Translate: $\mathbb{E}[X+Y] = 7$ always (linearity). $\mathbb{E}[XY]$ cannot be found without independence, so enter -1 if told to.

Caveat: Compare with A1. The only difference is whether independence is stated.

B. Geometric and Binomial stories (this is where the real Q2 was)

B1. Find p from a game

Each turn you roll two dice. You advance if they match or add to 7. T = number of turns up to and including the first advance. (a) Name the distribution and p. (b) $\mathbb{P}$(no advance in the first 2 turns). (c) $\mathbb{P}(T = 4)$. (d) $\mathbb{E}[T]$ and $\mathrm{Var}(T)$. (e) A = number of advances in 10 turns; name its distribution.

Translate: Matches: 6/36. Sums to 7: 6/36. They don't overlap (matches have even sums), so $p = 12/36 = 1/3$.
(a) $T \sim \mathrm{Geom}(1/3)$. (b) $(2/3)^2 = 4/9 \approx \mathbf{0.44}$. (c) $(2/3)^3(1/3) = 8/81 \approx 0.099$. (d) $\mathbb{E}[T] = 3$, $\mathrm{Var}(T) = \frac{2/3}{1/9} = 6$. (e) $A \sim \mathrm{Bin}(10, 1/3)$.

Caveat: Count the outcomes that make a success *first*, check for overlap, then pick the model.

B2. Geometric tail vs. exact

Roll a die until the first six. (a) $\mathbb{P}$(takes more than 4 rolls). (b) $\mathbb{P}$(first six is on roll 4). (c) $\mathbb{P}$(at least 4 rolls).

Translate: $p = 1/6$. (a) $\mathbb{P}(X > 4) = (5/6)^4 \approx 0.482$. (b) $(5/6)^3(1/6) \approx 0.096$. (c) $X \geq 4$ means the first 3 rolls fail: $(5/6)^3 \approx 0.579$.

Caveat: "More than 4" uses exponent 4, "at least 4" uses exponent 3.

B3. Memoryless [catch]

No six yet in the first 3 rolls. (a) $\mathbb{P}$(need at least 2 more rolls). (b) Expected total number of rolls.

Translate: (a) The future is a fresh Geom(1/6): $\mathbb{P}(X \geq 2) = \mathbb{P}(X > 1) = 5/6$. (b) $3 + 6 = 9$.

Caveat: The wait restarts, but the total still includes the rolls already gone.

B4. Binomial: at least one vs. a specific order

12 components each fail independently w.p. 0.05. (a) $\mathbb{P}$(at least one fails). (b) $\mathbb{P}$(exactly the first 2 fail and the rest work).

Translate: (a) $1 - 0.95^{12} \approx 0.460$. (b) One specific sequence, so no coefficient: $0.05^2 \cdot 0.95^{10} \approx 0.0015$.

Caveat: The binomial coefficient counts arrangements; if the problem fixes which trials succeed, there is none.

B5. Reuse p

The probability you advance in one turn is $p = 1/3$ (B1). Over 20 turns, $\mathbb{P}(A \geq 1)$? Over repeated games, how many games until the first one that lasts more than 3 turns?

Translate: $1 - (2/3)^{20}$. A game lasts more than 3 turns w.p. $(2/3)^3 = 8/27$, so games until the first such game $\sim$ Geom$(8/27)$.

C. Poisson exact expression + Chebyshev or Markov (this is where the real Q3 was)

C1. The real Q3, with the trap spelled out [catch]

Arrivals are Poisson with rate 15 per hour; X = arrivals in one hour. (a) Write an exact expression for $\mathbb{P}$(more than 20). (b) Suppose you only know $\mathbb{E}[X] = 15$ and $\text{Var}(X) = 15$. Chebyshev bound on $\mathbb{P}$(more than 20).

Translate: (a) $X > 20$ is $X \geq 21$, so

$$\mathbb{P}(X > 20) = 1 - \sum_{k=0}^{20} e^{-15} \frac{15^k}{k!}.$$

The sum starts at 0, stops at 20 (the largest value *not* wanted), and has e^{-15}.
(b) $X \geq 21$ means $X - 15 \geq 6$, so the event is inside $|X - 15| \geq 6$. Bound $= 15/6^2 = 15/36 \approx$ **0.4167**.

Caveat: Using $k = 5$ (the "20" in the problem) gives 0.6. That is the wrong threshold: the integer event starts at 21.

C2. "At least" vs. "more than"

Same setup as C1. Bound $\mathbb{P}(X \geq 20)$ and $\mathbb{P}(X > 20)$, then compare with the Markov bound for $\mathbb{P}(X \geq 21)$.

Translate: $X \geq 20$: $k = 5$, bound $15/25 = 0.6$. $X > 20$: $k = 6$, bound $15/36 \approx 0.417$. Markov: $15/21 \approx 0.714$.

Caveat: Markov only needs the mean and $X \geq 0$; Chebyshev uses the variance too, so it is usually tighter here.

C3. Time conversion

Calls arrive at rate 6 per hour. (a) Exact expression for $\mathbb{P}$(more than 5 calls in 30 minutes). (b) Chebyshev bound on the same event using only the mean and variance.

Translate: $\mu = 6 \cdot \frac{1}{2} = 3$. (a) $1 - \sum_{k=0}^{5} e^{-3} \frac{3^k}{k!}$. (b) $X \geq 6$, so $|X - 3| \geq 3$, bound $= 3/9 =$ **1/3**.

Caveat: Convert to the window length first, and don't forget Var $= \mu$ for Poisson.

C4. Two-sided Chebyshev

Mean 10, SD 2. (a) Bound $\mathbb{P}(X \geq 16)$. (b) Smallest guaranteed percentage of the distribution between 6 and 14.

Translate: (a) $X \geq 16$ is inside $|X - 10| \geq 6$, so the bound is $\leq 4/36 = 1/9$. (b) $k = 4$: $\mathbb{P}(|X - 10| < 4) \geq 1 - 4/16 = 75\%$.

Caveat: It is a bound, never an exact probability, and it covers both tails.

C5. Irrelevant condition

Rate 4 per hour. Given 7 calls came in the first hour, $\mathbb{P}$(at least one call in the next hour)?

Translate: Non-overlapping windows are independent, so the condition is irrelevant: $1 - e^{-4}$.

D. Unseen formats (backwards and combined)

D1. Geometric, backwards

$X \sim \text{Geom}(p)$ and $\mathbb{P}(X > 3) = 0.125$. Find p and $\mathbb{E}[X]$.

Answer: $(1-p)^3 = 0.125$, so $p = 0.5$ and $\mathbb{E}[X] = 2$.

D2. Poisson, equal neighbours

$X \sim \text{Poisson}(\mu)$ with $\mathbb{P}(X = 2) = \mathbb{P}(X = 3)$. Find μ.

Answer: $\mathbb{P}(3)/\mathbb{P}(2) = \mu/3 = 1$, so $\mu = 3$.

D3. Chebyshev, backwards

How many SDs k guarantee at least 96% of the distribution lies within k SDs of the mean?

Answer: $1 - 1/k^2 \geq 0.96$, so $k \geq 5$.

D4. Second moments

$X \sim \text{Poisson}(3)$. Find $\mathbb{E}[X^2]$ and $\mathbb{E}[(X-1)^2]$.

Answer: $\mathbb{E}[X^2] = 3 + 9 = 12$. $\mathbb{E}[(X-1)^2] = \text{Var} + (\mathbb{E} - 1)^2 = 3 + 4 = 7$.

D5. Geometric second moment

$X \sim \text{Geom}(1/4)$ and $Y \sim \text{Poisson}(2)$, independent. Find $\text{Var}(XY)$.

Answer: $\mathbb{E}[X] = 4$, $\text{Var}(X) = 12$, $\mathbb{E}[X^2] = 28$; $\mathbb{E}[Y^2] = 6$. $\text{Var}(XY) = 28 \cdot 6 - (4 \cdot 2)^2 = 168 - 64 = 104$.

D6. Markov for "at least one"

40 independent parts each fail w.p. 0.01; X = number failing. Markov bound on $\mathbb{P}(X \geq 1)$? Exact value?

Answer: $\mathbb{E}[X] = 0.4$, so $\mathbb{P}(X \geq 1) \leq 0.4$. Exact: $1 - 0.99^{40} \approx 0.331$. The bound is valid but loose.

5. Quiz-day checklist

- ☐ For "more than / fewer than / at least / at most" on an integer variable, write the integer event *first*.
- ☐ For Chebyshev, write $|X - \text{mean}| \geq k$ explicitly and double-check k.
- ☐ For Poisson, convert the time window first, then $\mu = \lambda t$.
- ☐ For Poisson sums, confirm the lower limit ($k = 0$), the upper limit, and the sign of the exponent.
- ☐ "Not enough information" is only for a *missing* fact. If independence is stated, you can use it.
- ☐ Before submitting multiple choice, reread the whole option, not just the shape.
- ☐ When rounding, compute the exact value, then round once at the end.